



S/he Bio on my Mimi 'til I Cry

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Field(s) of Interest: Engineering, Biology, Environmental Science

Brief Overview (1-3 sentences):

Students will learn about biomimicry, the practice of using ideas from nature to inspire human designs and solutions. This lesson will feature a famous example from Leonardo Da Vinci and students will model Da Vinci's inspiration from nature, understand how geckos stick on walls without any glue, and understand how engineers take inspiration from birds to build a variety of planes that each have their own unique abilities.

Agenda:

- Introduction (5 min)
- Module 1: Drawing like Da Vinci (10-15 min)
- Module 2: I like my lesson grippy bruh: Gecko Grip (15-20 min)
- Module 3: DIY Flock Camera: Birds (15-20 min)
- Conclusion (5 min)

Main Teaching Goals/Key Terms:

- Inspiration
- Design
- Engineering Design Process
- Van Der Waals forces
- Setae
- Spatulae
- Nanoscale Contact
- Adaptation
- Lift
- Drag
- Aerodynamics

Background for Mentors

Module 1

- Inspiration
- Design
- Engineering Design Process

Inspiration is the process of using an existing idea, object, organism, or observation as a starting point for creating something new. In biomimicry, engineers and inventors can observe how organisms solve problems in nature and use those observations as inspiration for human-made designs.

Leonardo da Vinci is an example of an inventor who closely observed birds and their wings and used his observations to inspire designs for human flying machines. He first took inspiration from observing the human body post-mortem and dissecting the human body for science.

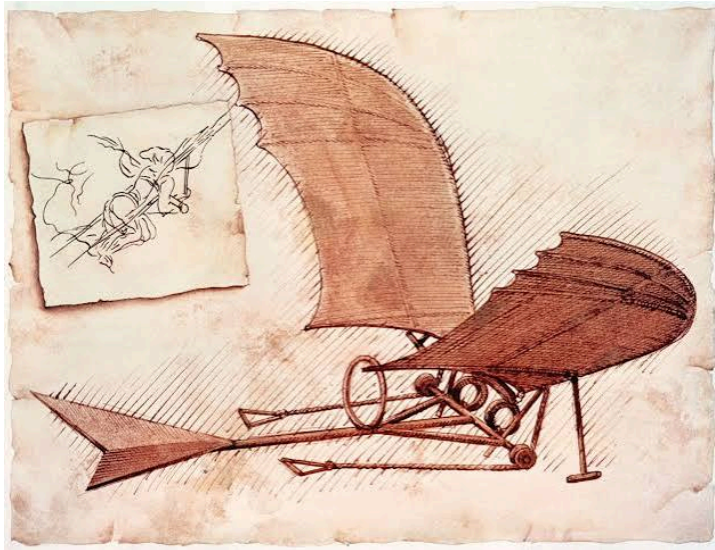


Figure 1: Leonardo Da Vinci's human flying model inspired by birds and their movement in ~1485

A **design** is a plan or representation of an idea that shows how something could look, work, or solve a problem. Engineers use designs to communicate their ideas before creating a physical product. Real-life example below:

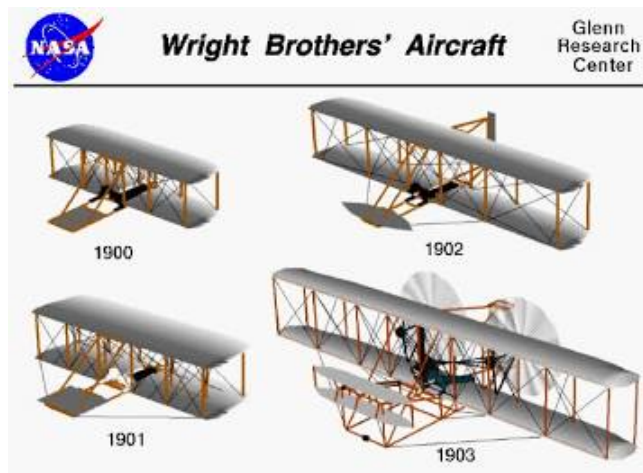


Figure 2: The Wright Flyer prototypes

The **engineering design process** is a series of steps engineers use to develop solutions to problems. Although the process can look different depending on the problem, it often involves identifying a problem, brainstorming ideas, designing a solution, building or creating a prototype, testing it, and improving the design.

In 1903, the *Wright Flyer* made the world's first controlled, sustained, and powered flight on December 17, 1903. This was not done on the first try, but rather through much trial and error as seen through the different prototypes created in *Figure 2*, before the final product in *Figure 3*.



Figure 3: The final version of the Wright Flyer in 1903

Module 2

- Setae
- Spatulae
- Nanoscale Contact
- Van der Waals forces

Geckos are able to climb smooth, vertical surfaces because of the specialized structures on the bottoms of their feet. Unlike many climbing animals, geckos do not rely on claws or sticky liquids to attach themselves to surfaces. Instead, their feet contain millions of microscopic structures that allow them to create extremely close contact with a surface.

Setae are microscopic, hair-like structures found on the bottom of a gecko's feet. A gecko has millions of setae across its toe pads. These structures increase the amount of surface area available for the gecko's foot to interact with a surface.

Spatulae are extremely small, flattened structures located at the ends of the setae. Because they are so small, spatulae can come into very close contact with the surface the gecko is climbing. This allows the gecko to create many points of contact between its feet and the surface.

Nanoscale contact refers to contact occurring at an extremely small scale, around one-billionth of a meter. The tiny size of gecko spatulae allows them to get extremely close to the surface, even when the surface appears smooth to us. This close contact is an important part of how gecko feet adhere to surfaces.

Van der Waals forces are weak attractive forces that occur between molecules when they are extremely close together. These forces can result from small fluctuations in the distribution of electrons within molecules, creating temporary attractions between neighboring molecules. A single van der Waals interaction is very weak, but the millions of tiny contact points created by a gecko's setae and spatulae allow these forces to add up to significant adhesion.

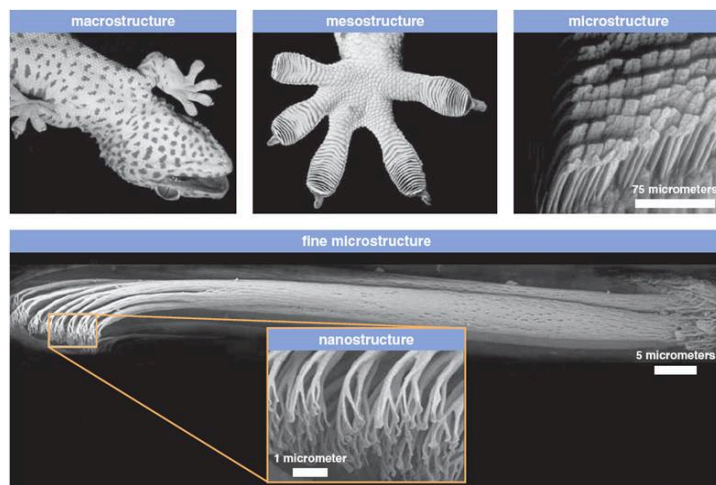


Figure 1: Macro and nano structure of gecko's toe

Module 3

- Adaptation
- Lift
- Drag
- Aerodynamics

Adaptation is a physical or behavioral trait that helps an organism survive in its environment. Birds have many adaptations that allow them to fly, and different birds have different adaptations depending on how they move through their environments. For example, birds that frequently soar, such as albatross, may have different wing shapes from birds adapted for fast or highly maneuverable flight, such as falcons. These differences can provide inspiration for designing different types of aircraft.

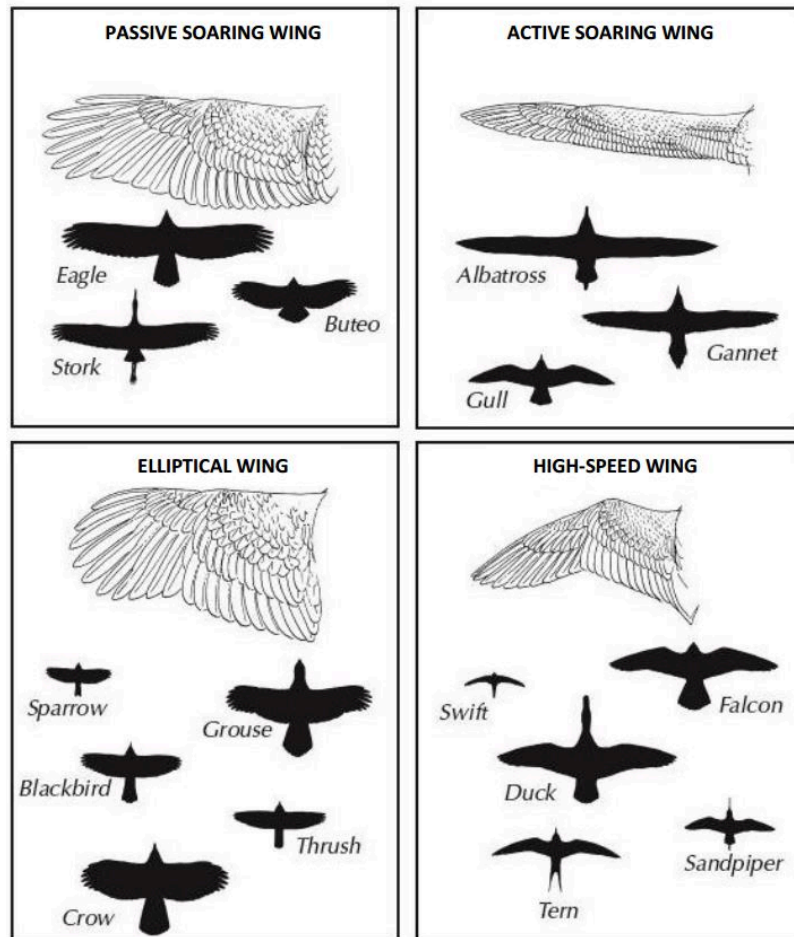


Figure 1: Bird wings after adaptation to their environment

When a bird or airplane moves through the air, it experiences different forces that affect how it flies. **Lift** is the force that acts upward and helps counteract gravity, allowing a bird or aircraft to stay in the air. As air moves past the moving wing, it splits. The shape and movement of its wings affect how air moves around them and how much lift is produced.. The air traveling over the curved top moves faster than the air underneath, creating lower air pressure above and higher pressure below. This pressure difference pushes the wing upward

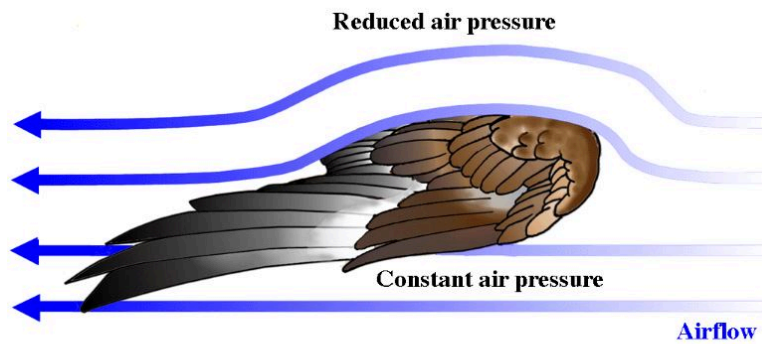


Figure 2: Bird wings being affected by lift

At the same time, **drag** is a force that resists an object's motion as it moves through the air. Birds use drag during takeoff by tilting their wings to use air resistance to help lift their body weight. Birds increase drag by spreading their wings and tails wide to slow down safely when landing. A bird's body shape and overlapping feathers are designed to minimize drag and make flight efficient

Aerodynamics is the study of how air interacts with moving objects. Engineers use principles of aerodynamics to design aircraft that move through air in specific ways.

Forces of Flight

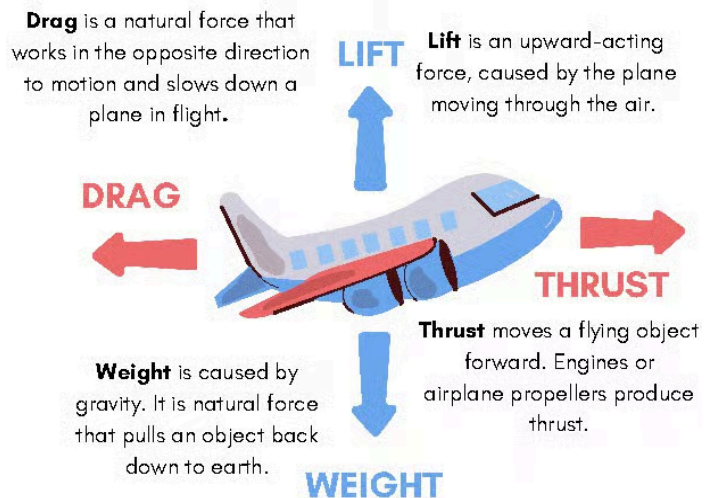


Figure 3: Aerodynamics core principles graph

Introduction

Concepts to Introduce • Biomimicry. "Bio" means life and "mimicry" means to imitate. So, biomimicry means to imitate life or nature. • Shinkansen Bullet Train — Redesigned with a nose mimicking a kingfisher's beak to eliminate loud tunnel sonic booms and cut energy use by 15% • Nanoscale: how there are things smaller than our eyes can see and mind can comprehend. ◦ Compare the drop in size using proportional analogies, such as comparing a meter to a nanometer the same way you compare the Earth to a marble. ◦ A human hair is roughly 80,000 to 100,000 nanometers thick. • Bumps on the flippers of humpback whales inspired quieter and more efficient turbine blades.	Questions to Pique Interest • What is something that animals are really good at doing? ◦ "Flying, climbing, swimming, etc" • Who here has a dog? Have you noticed that when you walk your dog outside, that sometimes prickly seed pods from weeds and plants latch onto your dog's fur? ◦ This inspired the invention of Velcro! You might see velcro on shoes and clothes to stick two things together. • Have you ever seen birds flying in a V-formation? ◦ Birds can save energy by creating an updraft for the ones behind them. Some scientists are considering having airplanes do the same thing to conserve fuel use.
Scientists, Current and Past Events • Leonardo da Vinci (1452–1519) was an artist, scientist, and inventor who closely observed the natural world. He studied birds, their wings, and their movements and created detailed sketches and designs for human flying machines inspired by what he observed. His work demonstrates the idea of observing nature to inspire engineering. ◦ Further Reading • The Wright Flyer, also known as the Kitty Hawk or 1903 Flyer, made the world's first controlled, powered, and sustained heavier-than-air human flight on December 17, 1903. It marked the beginning of the pioneer era of aviation.	Careers and Applications • Materials Scientist/Engineer: Creates advanced, lightweight, or eco-efficient materials modeled on natural structures like spider silk, bone, or plant surfaces for use in construction, aerospace, and consumer goods. • Environmental Engineer: Applies natural systems thinking to civil infrastructure, green buildings, and eco-friendly manufacturing. • Research Scientist / Biologist: Conducts fundamental research on living organisms in academic, lab, or corporate settings to uncover new mechanisms that engineers and designers can adapt.

Module 1: Drawing like Da Vinci

Mentees will learn how Leonardo da Vinci studied birds and sketched their wings as inspiration for designing human flying machines. Mentees will then choose an animal, plant, or other natural feature that interests them. **Working individually, mentees will design and sketch their own biomimicry invention inspired by nature.**

Teaching Goals 1. Inspiration: Using an idea, object, or process as a starting point for creating something new. 2. Design: Creating a plan or drawing for how something will work. 3. Engineering Design Process: Identifying a problem, brainstorming solutions, designing a solution, and improving the design.	Materials • One piece of paper per mentee (antique-looking paper). • One pencil per mentee to sketch their design • Any other coloring tools the mentees already have available
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Different Methods for Teaching

1. **Think-pair-share: brainstorm favorite animals!**
 - a. Get mentees to talk about what they like about animals and what unique features they have.
 - b. This is also a good time to walk around and get to know the mentees!
 - c. If mentees are having a hard time thinking of something, working with a partner might help.
2. **Da Vinci and the engineering design process**
 - a. Emphasize iterative
 - b. Visual print out of Da Vinci and the first iteration - relate to modern planes to see how far something can come from initial idea
3. **Use the whiteboard!**
 - a. Have a mentor write these stimulating questions down on a whiteboard (or similar) for mentees to think and answer through their drawing. Give an example of Da Vinci and the human flying machine → airplanes!
 - i. My organism
 - ii. Its special feature
 - iii. What problem does it solve?

Procedure

- Hand out 1 piece of paper for each mentee
- Have each mentee choose an animal, plant, or natural feature that they find interesting.
 - For younger mentees or mentees who still might be shy, do the Think-Pair-Share to get them to think and collaborate on an invention. But you should do this anyway!
- Encourage students to choose something based on a special ability (camouflage, nocturnal, etc). And how this ability might solve a problem humans or earth faces.
- After the children answer these questions, have them sketch out their invention on their paper.
- Encourage them to label the part of their invention that was inspired by nature.
- Have ~2-3 mentees stand up and present their inventions to the class.

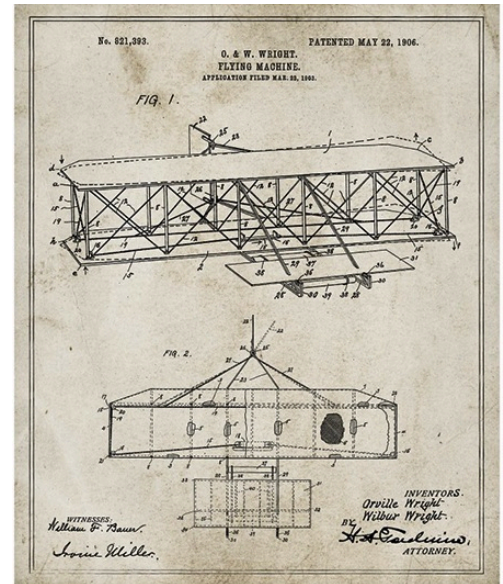


Figure 1: Original sketch on vintage paper of the Wright Flight

Classroom Notes

This is a good time to ease into conversation with mentees and get to know their interests, don't be shy and talk to them!

Module 2: I like my lesson grippy bruh (Gecko Grip)

Mentees will explore how geckos stick to surfaces through a short balloon demonstration and an individual spoon-gecko build. **They will learn how setae, spatulae, nanoscale contact, and Van der Waals forces help geckos climb, then use gecko feet as inspiration to create their own gripping geckos.**

Teaching Goals 1. Satae: Microscopic, hair-like structures found on the feet of certain animals. 2. Spetulae: Tiny, triangular split ends at the tip of each seta. 3. Van der Waals forces: Molecular attractions that operate over very small distances. 4. Nanoscale Contact: A physical connection or touch that occurs at the scale of nanometers (billionths of a meter). a. At this level, the distance between two objects is small enough for molecular attraction to take over.	Materials • 1 balloon • At least 4 sheets of felt for mentors. ◦ Mentors will hold this and mentees will use this to test their gecko's grip • A flat surface ◦ A wall or whiteboard works! Per mentee: • One wooden spoon • 3 pieces of green pipe cleaner • 2 small googly eyes • Scissors
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Different Methods for Teaching 1. For each teaching goal, have a visual! 2. Setae: tiny, hair-like structures on the bottom of a gecko's feet (Geckos have millions of setae across their toe pads). They increase the amount of surface area that can interact with a surface. a. Analogy: Think of a paintbrush. i. Paintbrush = lots of bristles // Gecko foot = lots of setae 3. Spatulae: tiny structures at the ends of the setae. They are extremely small and allow the gecko's foot to get very close to a surface. a. Analogy/Draw it out: "Zoom" in on one bristle of a hairbrush and imagine its end becoming even smaller and more specialized. 4. Nanoscale Contact: Imagine having a microscope that lets you zoom in way beyond what your eyes can see. Even surfaces that look completely smooth to us have tiny bumps and spaces when viewed at a very small scale (analogy). 5. Van der Waals forces are weak molecular attractions that occur when molecules are extremely close together. One interaction is very weak, but a gecko has millions of setae and spatulae, creating millions of tiny contact points.

- a. Analogy: Think of one tiny attraction as weak, but millions of tiny attractions working together as much stronger. Such as ants working together!

Part 1: Demo

1. Select 1 or 2 mentees for a class demonstration (call them to the front).
 - a. **Ensure they aren't allergic to latex**
2. Give each selected mentee an inflated and tied balloon.
3. Have the mentees rub their inflated balloons on their hair for approximately 10 seconds.
4. Immediately after rubbing the balloons, have the mentees gently press their balloons against the wall and *slowly* let go. See **Figure 1**
5. Observe whether the balloons remain attached to the wall.
 - a. Ask: "How is the balloon able to stay on the wall without glue?"
 - b. CONNECTION: Van der Waals forces (specialized structures on their feet)

Part 2: Grippy

1. Now their challenge is to design and build their own gecko that can stick to surfaces without needing glue.
2. Pass out materials Per mentee:
 - a. Scissors to share
 - b. 1 wooden spoon.
 - c. 2 pipe cleaners
 - d. 2 googly eyes for joy and divine creation.

Gecko Making and Testing:

3. Fold three pipe cleaners in half.
4. Wrap one around the top end of the wooden spoon to create the front legs (**Figure 2**) and one around the bottom end to make the back legs (**Figure 3**).
 - a. If too long, mentees can cut the legs
5. Wind the tail onto the pipe cleaner legs at the back on top of the spoon. See **Figure 3**.
6. Attach googly eyes to make the gecko's face



Figure 1: Image of how the balloon should stick to the wall with no adhesive.



Figure 2: Close up of how the pipe cleaners will be wrapped to make the front legs.

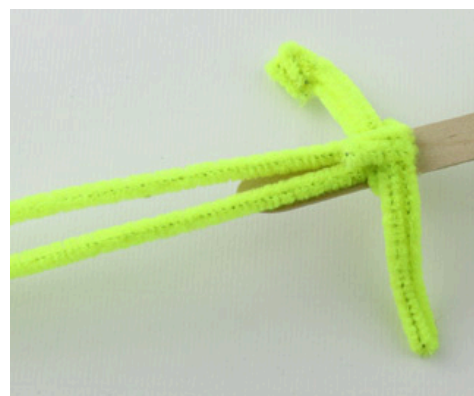


Figure 3: Close up of how the pipe cleaner will be wrapped to make the back legs and tail.

7. Have mentors walk around with the felt for mentees to test their gecko.
 - a. If having trouble, tilt the fabric 45° towards you to make the fabric angled where their gecko will for sure stick.

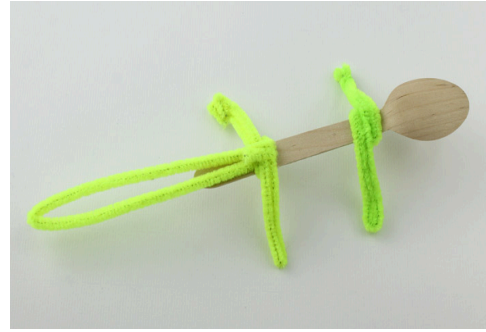


Figure 4: Image of gecko's finished body without the googly eyes

Classroom Notes

- Mentees can keep their geckos
- If the balloon activity doesn't work for any reason, it's okay! Just move on to the gecko activity and say "we'll see these forces work with our own geckos!"

Module 3: DIY Flock Cameras: Birds

Mentees will investigate how different bird wing shapes can inspire aircraft designs by building and testing four different paper airplane prototypes. Mentees will then choose one out of four designs inspired by birds to create their own aircraft.

Teaching Goals

1. **Adaptation:** A physical or behavioral trait that helps an organism survive and reproduce in its environment.
2. **Lift:** A force that helps keep an object in the air.
3. **Drag:** A force that opposes an object's motion through air.
4. **Aerodynamics:** The study of how air interacts with moving objects.

Materials

- 1 piece of standard copy paper for each student in the classroom. (May be ~35 pieces of paper)
- 5 copies of the four paper airplane designs (one per mentor).
- 15 copies of the four paper airplane designs for students to follow
- Crayons and markers for students to decorate their aircraft
- Scissors for students to edit their design

Different Methods for Teaching

1. **Adaptation:** Draw on the board birds and different wing shapes and sizes.
 - a. **Think-Pair-Share** on: “Why do you think these birds might have different wing shapes?”
 - b. Guide mentees toward the idea that birds have different adaptations for different types of flight. No single “best” wing shape. Different designs work better for different purposes.
2. **Lift Analogy:** ask, “Have you ever put your hand out of a car window?” And ask them to imagine their hand flat vs tilted. Which one causes less resistance from the wind?
 - a. Flat hand: less upward movement
 - b. Tilted hand: air pushes your hand upward
 - c. This is the same for how airplane wings use air to create lift. Easier to maneuver!
3. **Drag Analogy:** “Have you tried walking when it’s really windy before? Maybe San Francisco or near the water? The wind pushes against you and makes it harder to move forward. That’s similar to drag!”
4. **Aerodynamics:** not every plane, bird, or anything is really the same. Aerodynamics is just the field that studies how different things react to air.

Procedure

1. Introduce the four aircraft designs that mentees will build and test (should have one of each design as they were the Decal Activity):
 - a. **The Falcon** (Figure 1): Inspired by fast-flying birds such as peregrine falcons.
 - b. **The Owl** (Figure 2): Inspired by owls and designed to investigate how a different body and wing shape affects flight.
 - c. **The Eagle** (Figure 3): Inspired by soaring birds such as eagles, with a broader wing design.
 - d. **The Albatross** (Figure 4): Inspired by long-distance soaring birds with large wings.
2. Group mentees into groups, one mentor will be the guide on how to fold the paper airplane
 - a. *Students will not get to choose the design, it is whatever the mentor is specializing in that the mentees will follow based on where the mentor goes.*
3. Give each mentee 1 standard copy paper for them to use to fold their aircraft design.
 - a. Encourage mentees to think about the shape and size of their wings and how these features might affect flight.
4. After they are done folding, they are free to decorate their bird-inspired aircraft.
5. *If given permission by the teacher in the classroom:* mentees who would like to test their aircraft may bring their aircraft into a spot deemed safe by the teacher. Mentors will go one-by-one testing how far each airplane goes.
 - a. **Have mentees test their aircrafts low to the ground to avoid danger!**

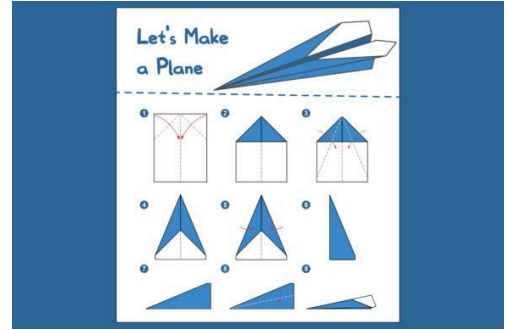


Figure 1: *The Falcon: Inspired by fast-flying birds such as peregrine falcons.*

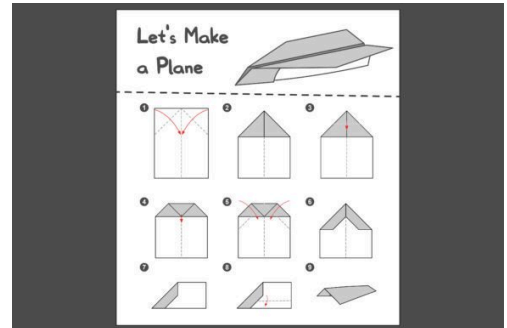


Figure 2: *The Owl: designed to investigate how a different body and wing shape affects flight.*

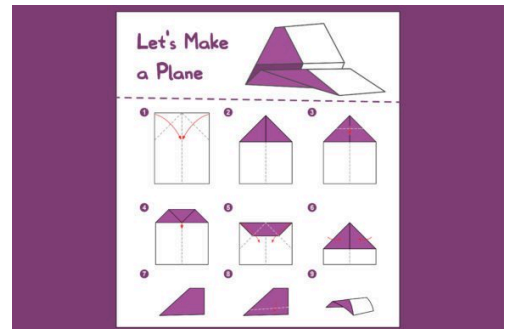


Figure 3: *The Eagle: Inspired by soaring birds such as eagles, with a broader wing design.*

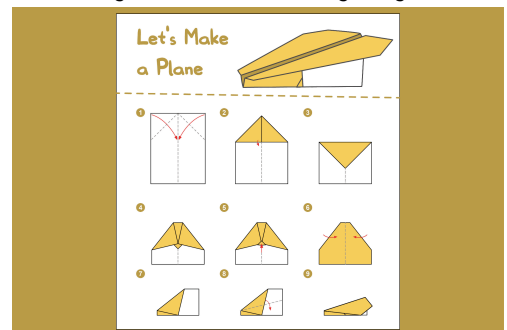


Figure 4: *The Albatross: Inspired by long-distance soaring birds with large wings.*

Classroom Notes

- For younger sites, prioritize completing and understanding one airplane rather than completing all four designs. Or, Mentors can show the Falcon, Owl, Eagle, and Albatross models made in Decal as examples and explain that different birds have different wing shapes for different types of flight.
- If permission is not given or for students who are finished before others: encourage them to design their aircrafts using crayons and markers.

Conclusion

1. Have mentees share one example of biomimicry from the lesson!
2. Animals and plants have developed structures and behaviors that help them survive. Humans can observe these natural solutions and use them as inspiration to solve our own problems.

References

- Gecko Activity Inspiration:
<https://www.pbs.org/parents/crafts-and-experiments/grip-like-a-gecko>

Summary Materials Table

Material	Amount per Site	Expected \$\$	Vendor (or online link)
White Printer Paper	1 per student	\$0.00	
Antique Paper	1 per student	\$6.99 (80 pcs)	Amazon
Pencils	1 per student	\$0.00	
Balloons	2 per site	\$6.99 (100 pcs)	Amazon
Felt	4-5 per site	\$10.68 (42 pcs)	Amazon
Wooden Spoons	1 per student	\$8.99 (100 pcs)	Amazon
Green pipe cleaners	2 per student	\$5.39 (200 pcs)	Amazon
Small googly eyes	2 per student	\$7.99 (1000 pcs)	Amazon
Scissors	1 per student	\$0.00	